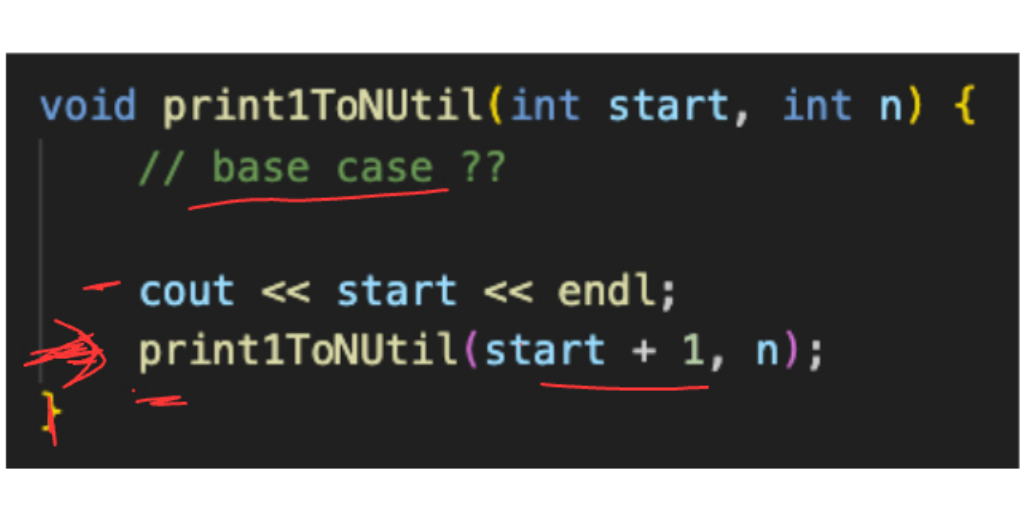
i = 1 → n
print (i)

Break the problem into smaller & similar
sub problems

print (1, n) → 1 "print (2, n)"



```
void print1ToNUtil(int start, int n) {
    // base case ??

    cout << start << endl;
    print1ToNUtil(start + 1, n);
}
```

n = 5

1
2
3
4
5

start = 1
start = 2
start = 3
start = 4
start = 5
start = 6

print (n) → n print (n-1)

Tail Recursion.

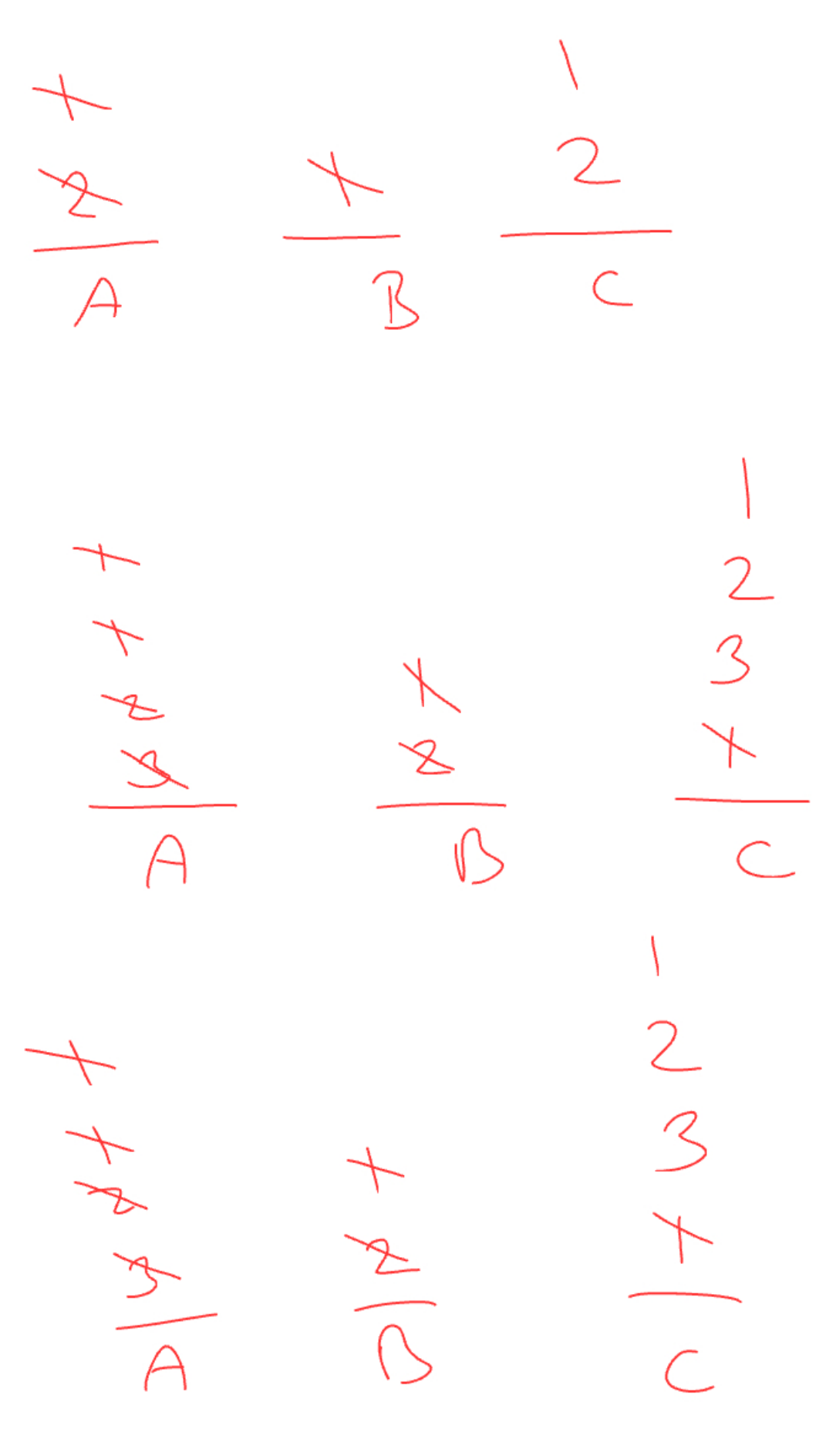
when the rec. call is the last thing your func does

fact (n) = n * fact (n-1)

Tower of Hanoi.

n=2.

|||

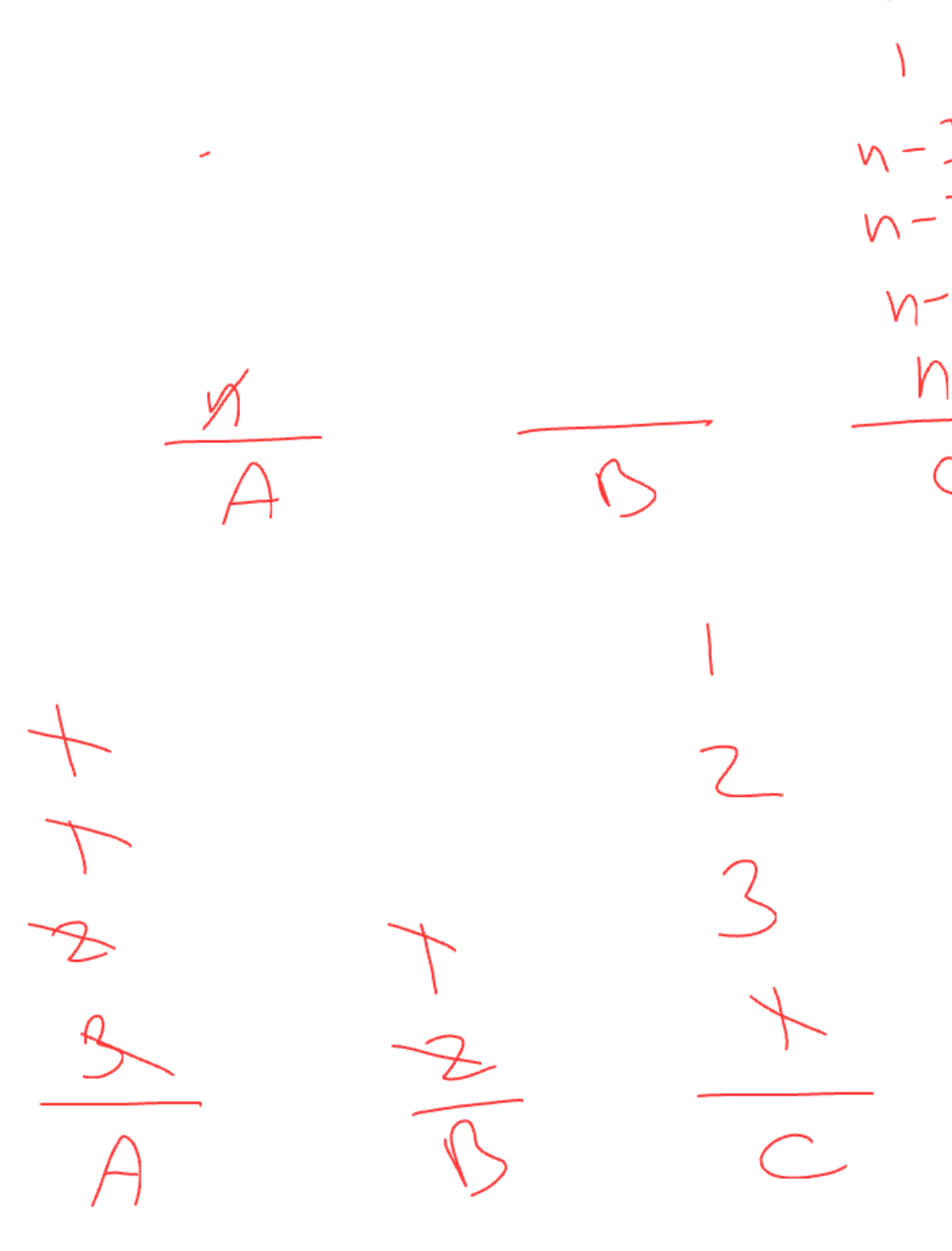


HHI ||

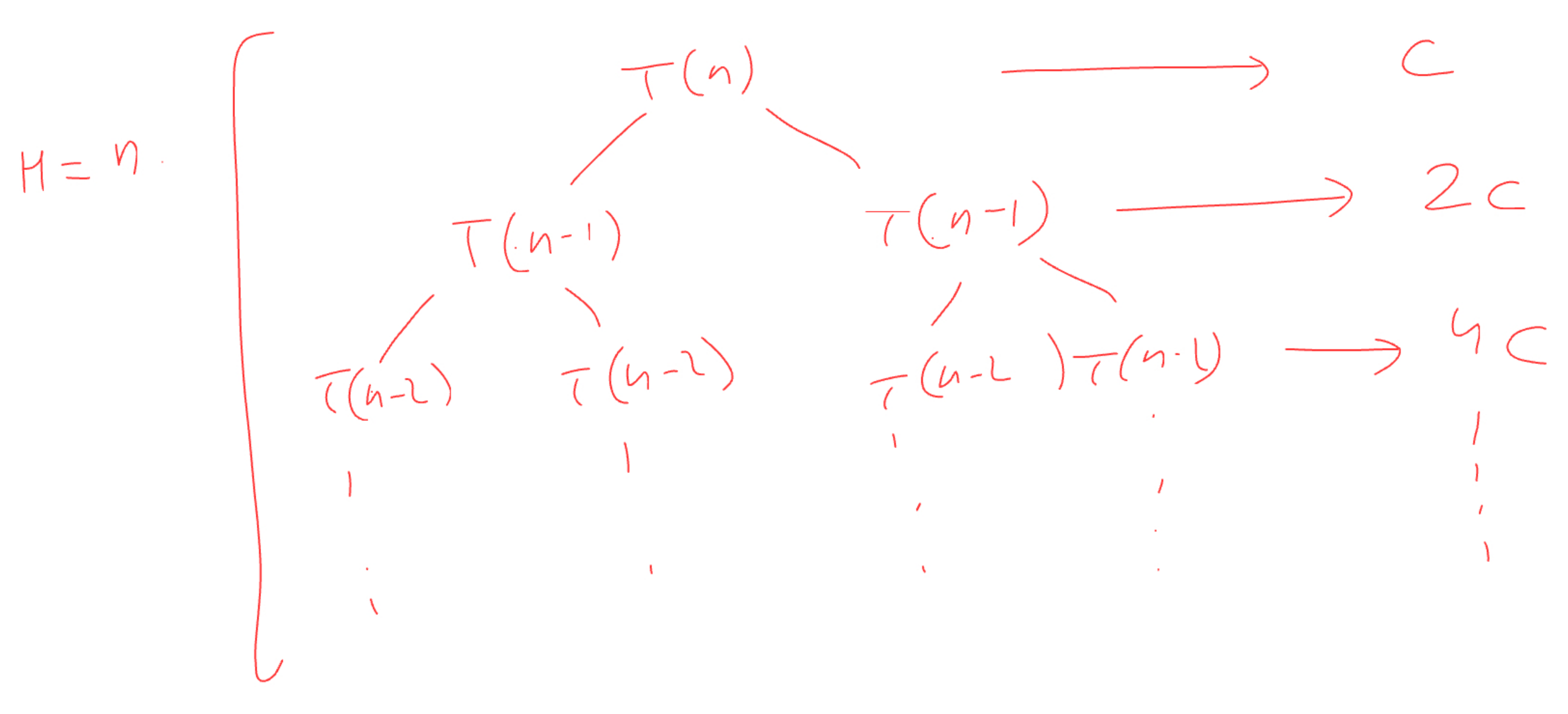
7.

(n, A, B, C) → (n-1, A, C, B) + ① + (n-1, B, A, C)

nth disk from
A to C


1
n-3
n-2
n-1
n
A
B
C
1
2
3

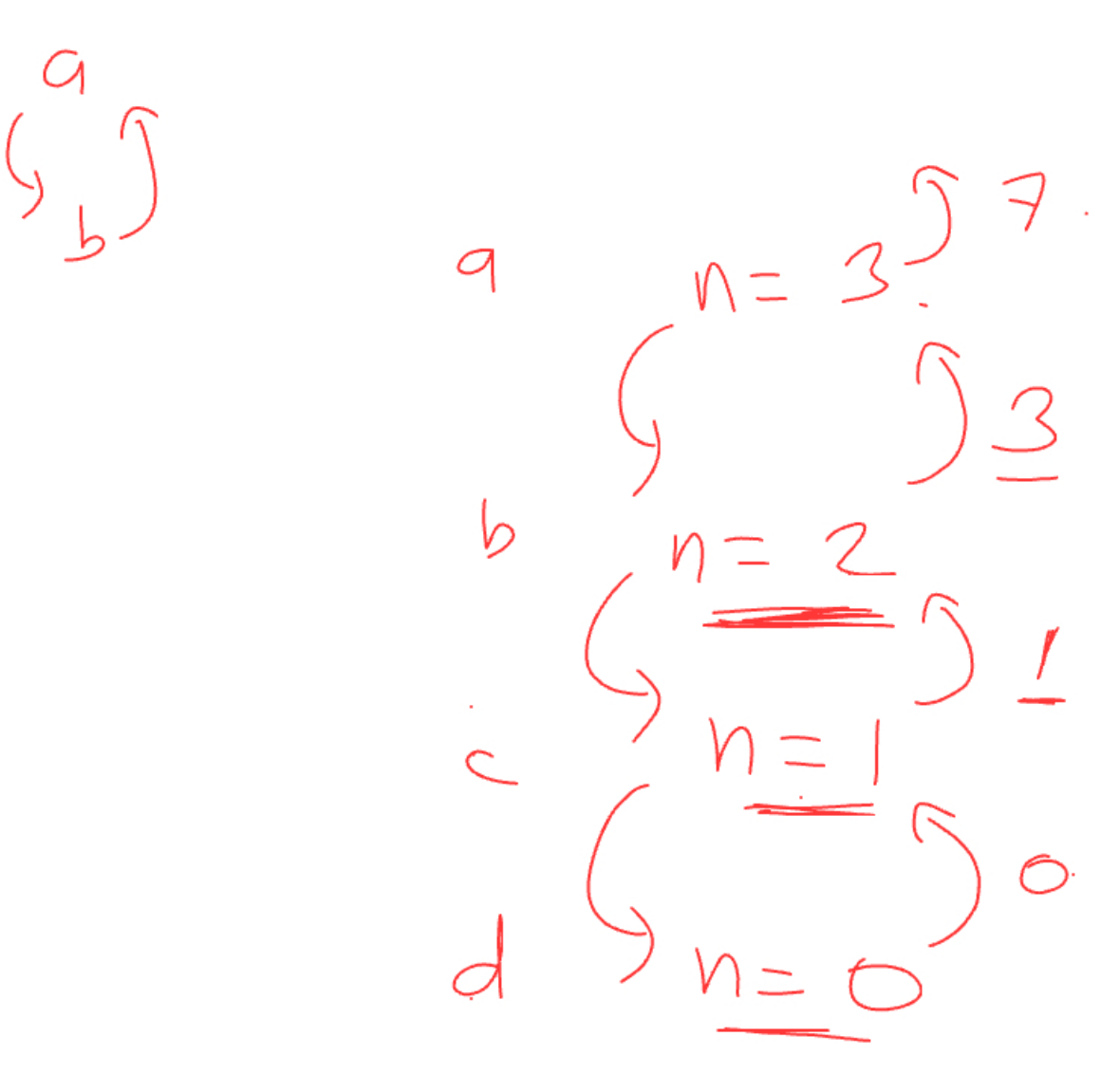
T (n) = 2 T(n-1) + C

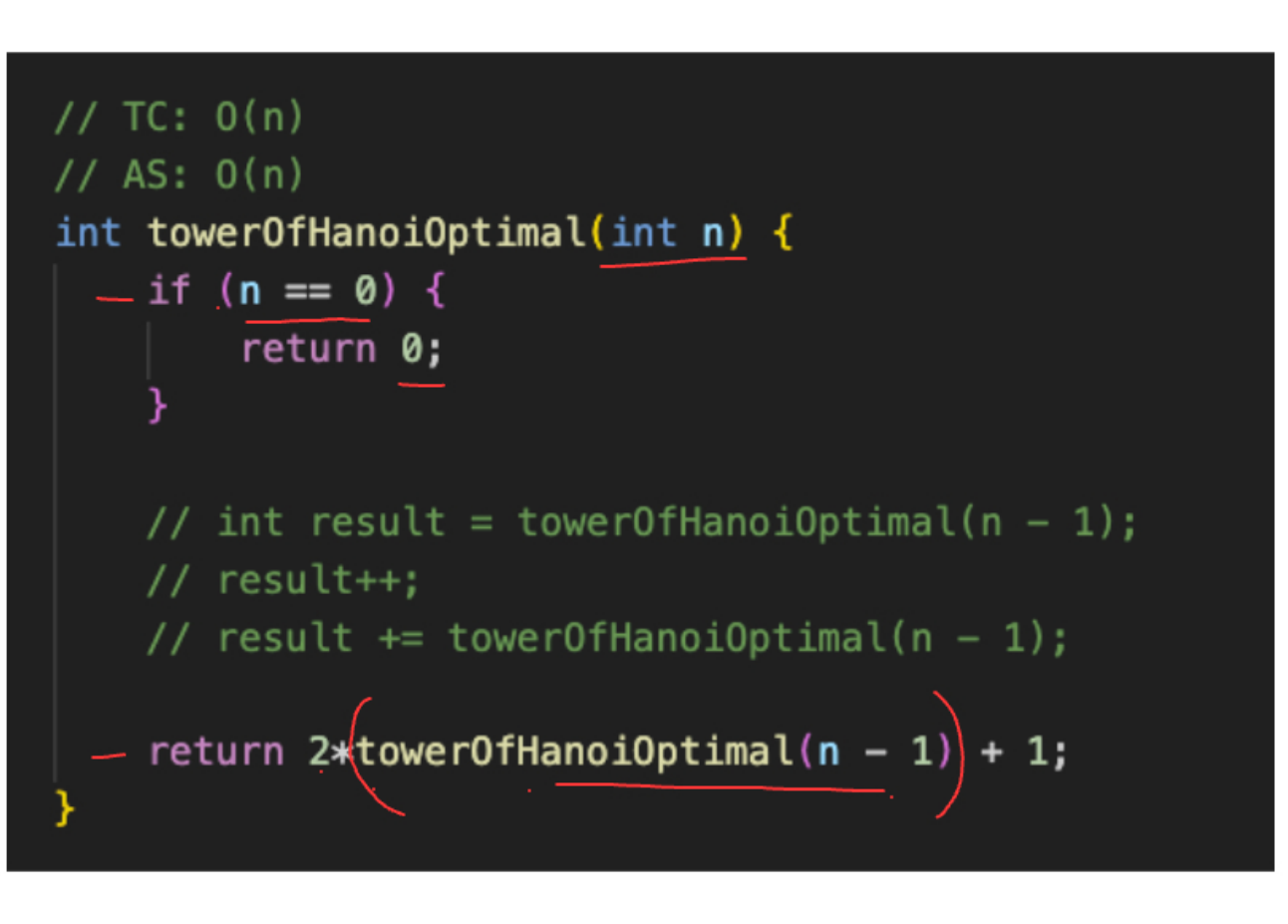

H = n
T(n)
T(n-1)
T(n-1)
T(n-2)
T(n-2)
T(n-2)
T(n-1)
C
2C
4C

$\underbrace{C + 2C + 4C + 8C - - - -}_{n\text{-terms.}}$

a = C
r = 2
n = n

$\frac{C(2^n - 1)}{2-1}$ = TC = $O(2^n)$


a
b
a
b
c
d
n = 3
n = 2
n = 1
n = 0
7
3
1
0



```
// TC: O(n)
// AS: O(n)
int towerOfHanoiOptimal(int n) {
    if (n == 0) {
        return 0;
    }

    // int result = towerOfHanoiOptimal(n - 1);
    // result++;
    // result += towerOfHanoiOptimal(n - 1);

    return 2*towerOfHanoiOptimal(n - 1) + 1;
}
```

Possible words from Phone Digits.

[2, 3]

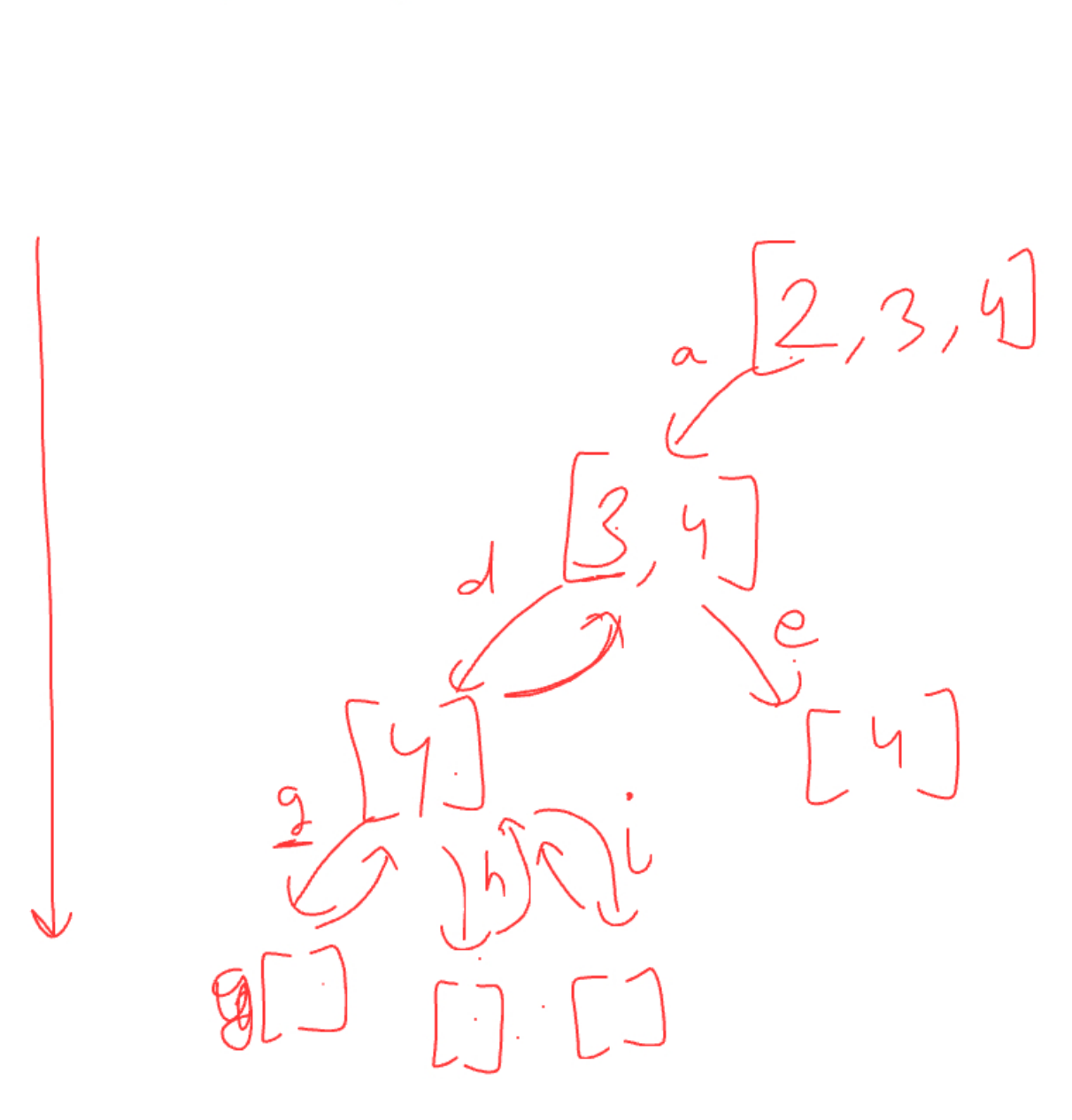
[2, 3, 4].

abc def ghi
[2, 3, 4] → [2, 3]

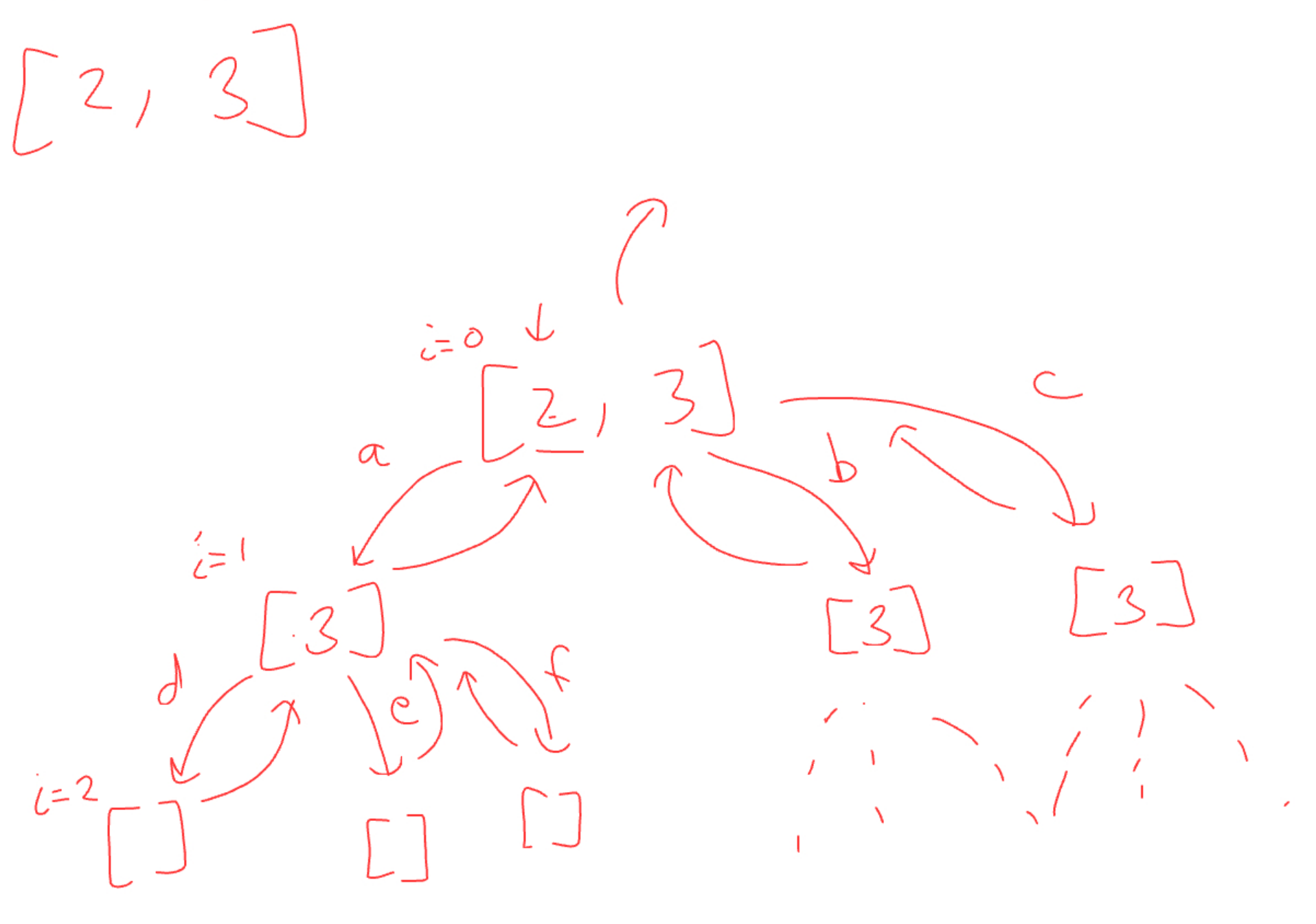
ad, ae, af
bd, be, bf
cd, ce, cf

abc def ghi
[2, 3, 4]

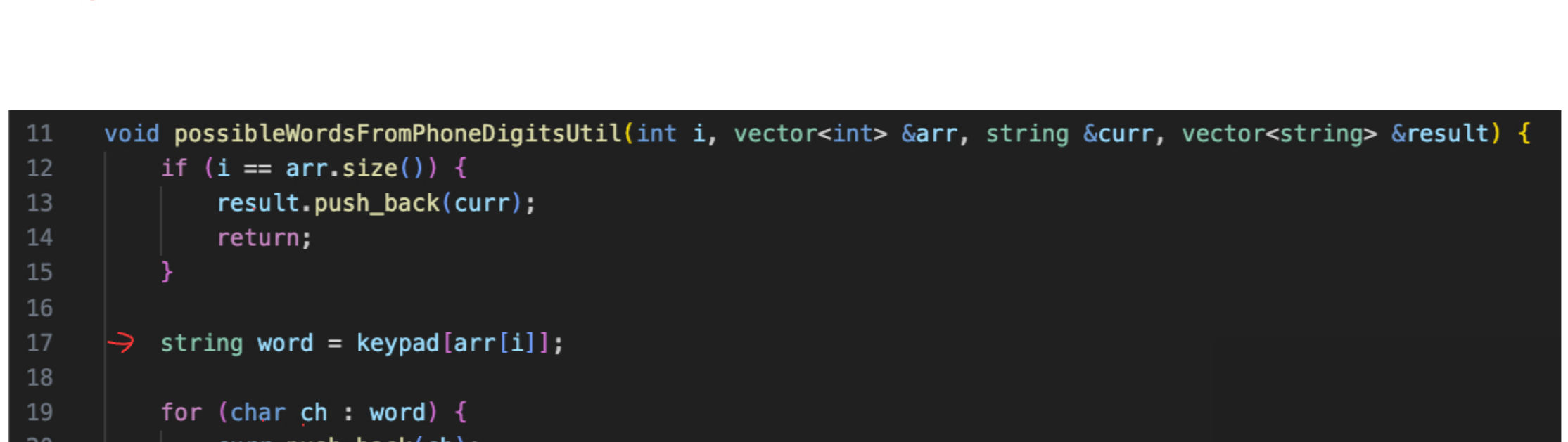
curr = ad~~ghi~~e
res = [adg, adh, adi
]


a
[2,3,4]
d
[3,4]
e
[4]
g
[4]
h
i
g[]
[]
[]

abc def
[2, 3]


i=0
[2, 3]
a
b
c
i=1
[3]
[3]
[3]
d
e
f
i=2
[]
[]
[]

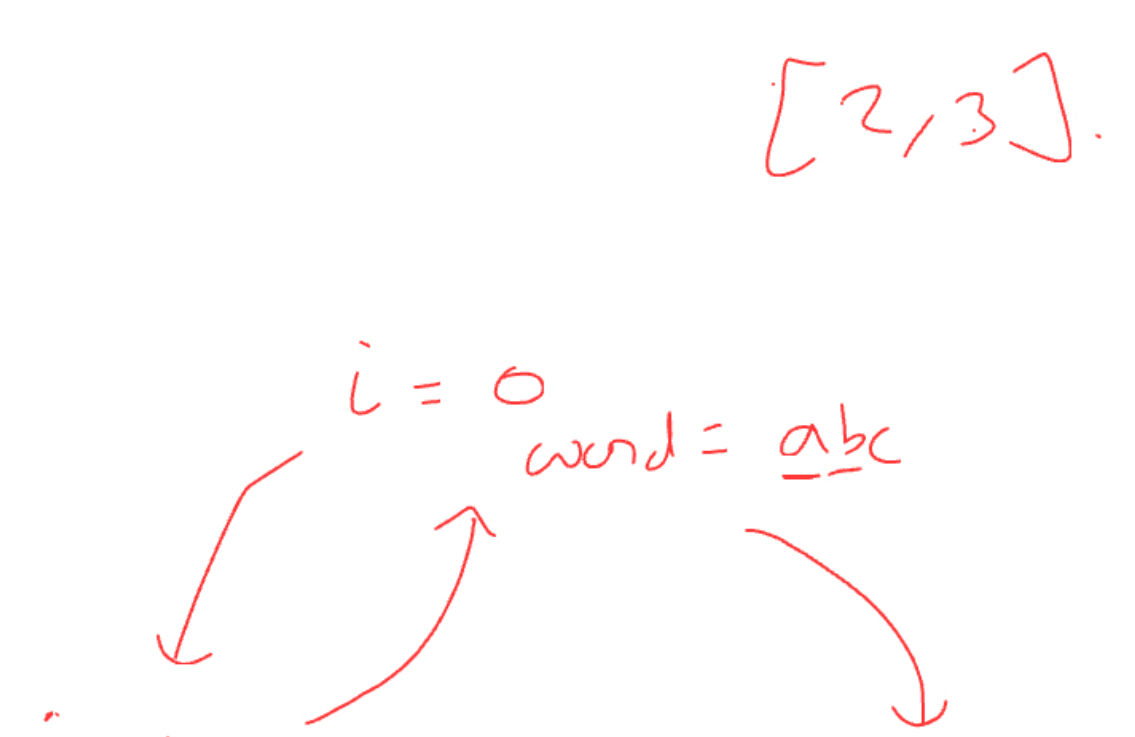
curr = ~~a~~ def ~~b~~
res = [ad, ae, af
bd, be, bf
cd, ce, cf

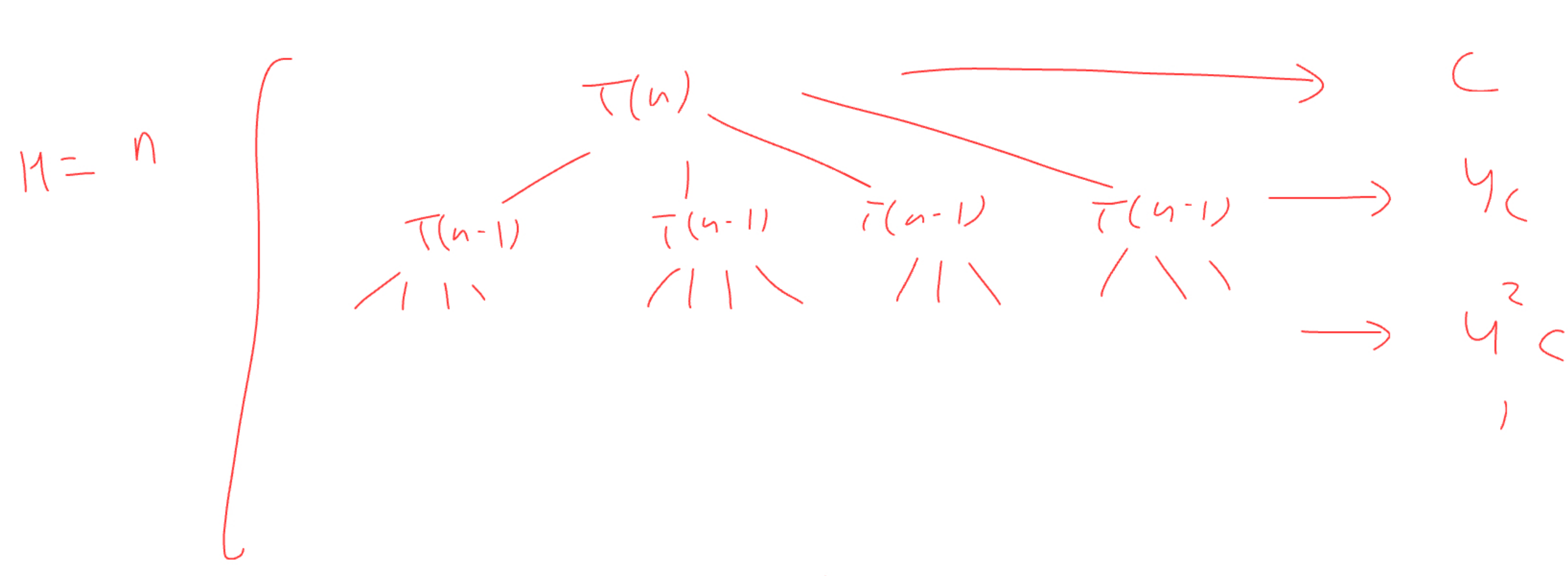


```
void possibleWordsFromPhoneDigitsUtil(int i, vector<int> &arr, string &curr, vector<string> &result) {
    if (i == arr.size()) {
        result.push_back(curr);
        return;
    }

    string word = keypad[arr[i]];

    for (char ch : word) {
        curr.push_back(ch);
        possibleWordsFromPhoneDigitsUtil(i + 1, arr, curr, result);
        curr.pop_back();
    }
}
```

[2,3].

curr: ~~a~~def b
res = [ad, ae, af


i = 0
word = abc
i = 1
word = def
i = 1
word = def
i = 2
i = 2
i = 2


H = n
T(n)
T(n-1)
T(n-1)
T(n-1)
T(n-1)
C
4C
4^2C

TC = $O(4^n)$.
AS =